

FI 362 — ADVANCED FINANCIAL ECONOMETRICS

Mock Midterm Examination |

Financial Time Series & Linear Time Series Analysis

Practice paper

Time allowed: 120 minutes

Total: 100 points

Part A: 5 multiple-choice ($5 \times 10 = 50$ pts) **Part B:** 1 solution/interpretation question (50 pts)

Instructions. Answer *all* questions. For Part A circle exactly one letter per question. For Part B show your reasoning: numerical answers with no interpretation earn little credit. Calculators permitted; no notes. Useful values: $1.96/\sqrt{T}$ for the ACF band; $\chi^2_{2,0.05} = 5.99$, $\chi^2_{10,0.05} = 18.31$; Dickey–Fuller 5% critical values -2.86 (with drift) and -1.94 (no drift).

This is a practice paper. It mirrors the format, coverage and difficulty of the graded midterm (Chapters 1–2 only); the numbers and scenarios are new.

Part A — Multiple Choice (10 points each)

- A1.** Let R_t be the simple net return and $r_t = \ln(1 + R_t)$ the log return of an asset, and let a portfolio hold weights w_i in assets i . Which statement is **true**?
- (A) Log returns aggregate *exactly* across the assets of a portfolio, $r_{p,t} = \sum_i w_i r_{i,t}$.
 - (B) Simple returns aggregate *additively* over time, $R_t(k) = R_t + R_{t-1} + \cdots + R_{t-k+1}$.
 - (C) Simple returns aggregate *exactly* across the assets of a portfolio, $R_{p,t} = \sum_i w_i R_{i,t}$.
 - (D) For any given price move the log return is always *larger* than the simple return.
- A2.** A daily return series has sample skewness ≈ -0.6 and sample *excess* kurtosis $\approx +6.5$. The most accurate reading is:
- (A) The distribution has *fat tails* and a *heavier left tail*: extreme moves — especially large *losses* — are more likely than a Normal with the same variance would predict.
 - (B) The returns are thinner-tailed than the Normal, so large moves are rare.
 - (C) The positive number (+6.5) is a skewness measure implying frequent large *gains*.
 - (D) Excess kurtosis of +6.5 equals the Normal benchmark, so normality holds.
- A3.** A stationary series has a sample ACF that *decays geometrically* (tails off), while its PACF is significant at lag 1 and negligible from lag 2 onward (cuts off). The most appropriate model is:
- (A) MA(1).
 - (B) ARMA(1,1).
 - (C) AR(2).
 - (D) AR(1).

- A4.** Consider the model $x_t = 0.5 + 0.9x_{t-1} + a_t$ with $a_t \sim \text{WN}(0, \sigma_a^2)$. Which statement is **true**?
- (A) It is non-stationary because the AR coefficient 0.9 is close to 1.
 - (B) It is weakly stationary with unconditional mean $\mathbb{E}[x_t] = 0.5/(1 - 0.9) = 5$, and its ACF decays geometrically as $\rho_k = 0.9^k$.
 - (C) Its long-run mean equals the intercept, $\mathbb{E}[x_t] = 0.5$.
 - (D) Its ACF cuts off to zero after lag 1.
- A5.** Two *independent* $I(1)$ series are regressed on each other by OLS. The output shows $R^2 = 0.92$ and a slope t -statistic of 210, but the residuals have a lag-1 autocorrelation of 0.98. This is best described as:
- (A) A genuine cointegrating relationship, confirmed by the high R^2 .
 - (B) Proof that the two series are white noise.
 - (C) Evidence that the OLS standard errors are *too large*, understating the true significance.
 - (D) A spurious regression: the OLS t -statistic is unreliable because the near-unit-root residuals make its standard error far too *small*; difference the data or use HAC (Newey–West) standard errors.

Part B — Solution & Interpretation (50 points)

An analyst studies $T = 1600$ daily log returns (in decimal) of an equity index ETF ABC. The computer output is summarised below.

Table 1 — Descriptive statistics of the daily log returns

mean	sd	skewness	excess kurtosis	Jarque–Bera	JB p -value
0.0003	0.011	−0.60	6.5	≈ 2913	< 0.001

Table 2 — Autocorrelation diagnostics

	lag-1 autocorrelation	Ljung–Box $Q(10)$ (p -value)
returns r_t	+0.062	31.5 (< 0.001)
squared returns r_t^2	+0.30 (and still > 0 at lag 150)	2600 (< 0.001)

Table 3 — Augmented Dickey–Fuller tests

series	ADF t -statistic	5% critical value
log <i>price</i> of ABC (with drift)	−1.85	−2.86
log <i>return</i> of ABC (no drift)	−28.7	−1.94

- (a) **(12 pts)** From Table 1 the Jarque–Bera statistic is $JB \approx 2913$.
- (i) Compare it with the critical value $\chi_{2,0.05}^2 = 5.99$: are the returns plausibly Normal? Reject or not?

- (ii) The excess kurtosis is +6.5. In one sentence, what does a large *positive* excess kurtosis say about the tails, and which stylised fact of returns is this?
 - (iii) The skewness is -0.60 . What does the *negative* sign say about up-moves versus down-moves?
- (b) **(12 pts)** With $T = 1600$ the ACF significance band is $\pm 1.96/\sqrt{1600} = \pm 0.049$.
- (i) Is the return's lag-1 autocorrelation of $+0.062$ inside or outside the band — i.e. is it statistically significant?
 - (ii) The one-step R^2 of an AR(1) equals ρ_1^2 . Compute it as a percentage. Is the predictable part of daily returns economically large or tiny?
 - (iii) Name one market feature (microstructure) that can create a small lag-1 autocorrelation in an *index* return series.
- (c) **(12 pts)** Read Table 3.
- (i) For the log *price*, do we reject the unit-root null? Is the price $I(0)$ or $I(1)$?
 - (ii) For the log *return*, do we reject? Is it $I(0)$ or $I(1)$?
 - (iii) In one sentence, why do we model *returns* rather than prices?
- (d) **(14 pts)** Compare the two rows of Table 2.
- (i) Which ACF dies out almost at once, and which stays large for hundreds of lags?
 - (ii) Is it the *direction* (mean) or the *size/volatility* (variance) of returns that is predictable? One sentence.
 - (iii) Which class of model is designed to capture the pattern in the *squared* returns (name the family)?

Hint / formulas. Reject Normality if $JB > \chi_{2,0.05}^2 = 5.99$. A correlation is significant if it falls *outside* the band $\pm 1.96/\sqrt{T}$. ADF: reject the unit root (conclude *stationary*) if the t -statistic is *below* (more negative than) the Dickey–Fuller critical value. The Normal distribution has excess kurtosis 0; positive excess kurtosis = fatter tails.